

Instructions

There are 13 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in blue or black pen. You may use pencil in graphs and diagrams only.

This examination booklet will be scanned and your work will be presented to an examiner on screen. Anything that you write outside of the answer areas may not be seen by the examiner.

Write your answers into this booklet. There is space for extra work at the back of the booklet. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

(Suggested maximum time: 5 minutes)

(i) $367 - 212$

[illegible][illegible][illegible]

0.14,	$\frac{1}{8},$	$\frac{17}{100},$	9%
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[illegible]

Answer = , , ,

Fill in the missing values in the table below to show how much of each ingredient is required to make each batch of cookies.

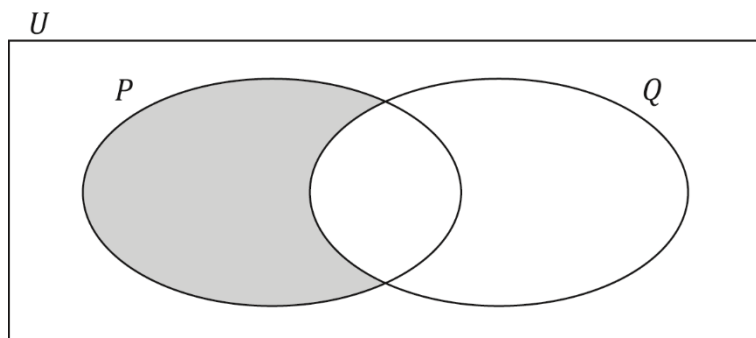
	6 Cookies	12 Cookies
Sugar	115 g	
Butter		460 g
Flour		

[illegible]

Question 2

(Suggested maximum time: 10 minutes)

- (a) Tick the box that correctly indicates the shaded region represented on the Venn diagram below. Tick (✓) **one** box only.

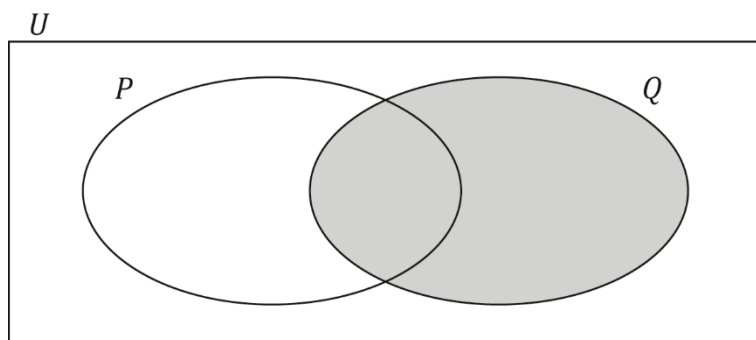


P
☐

$P \cap Q$
☐

$P \setminus Q$
☐

- (b) Tick the box that correctly indicates the shaded region represented on the Venn diagram below. Tick (✓) **one** box only.

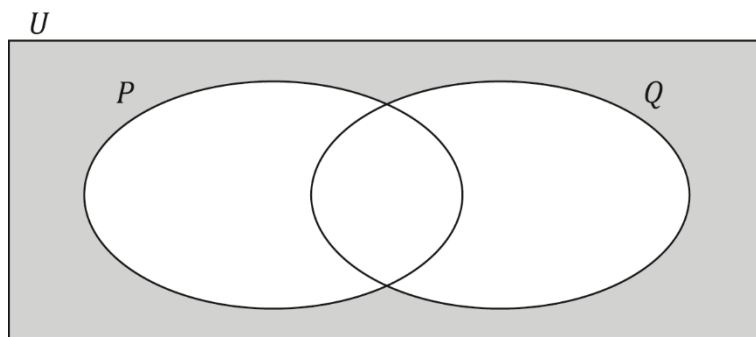


$P \setminus Q$
☐

Q
☐

$\#Q$
☐

- (c) Tick the box that correctly indicates the shaded region represented on the Venn diagram below. Tick (✓) **one** box only.



$P \cap Q$
☐

$(P \cup Q)'$
☐

$P \cup Q$
☐

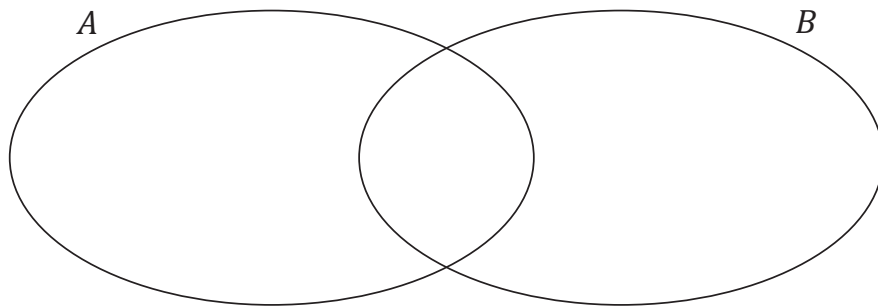
Note: $(P \cup Q)'$ is the complement of the set $(P \cup Q)$.

(d) The sets A and B are defined as follows:

$$A = \{J, O, S, E, P, H\}$$

$$B = \{P, A, U, L, I, N, E\}$$

(i) Complete the Venn diagram below to show the sets A and B .



(ii) Use your Venn diagram to list the elements of $A \cap B$.

$$A \cap B = \{ \quad \quad \quad \}$$

(iii) Use your Venn diagram to list the elements of $A \setminus B$.

$$A \setminus B = \{ \quad \quad \quad \}$$

(iv) Write down $\#(A \setminus B)$.

$$\#(A \setminus B) = \boxed{\quad \quad \quad}$$

(v) The set C is a subset of $A \setminus B$ and $\#C = 2$. Write down an example of the set C .

$$C = \{ \quad \quad \quad \}$$

Question 3 (Suggested maximum time: 15 minutes)

The fare is calculated as follows:

Each additional kilometre costs €1.50 for the remainder of the journey.

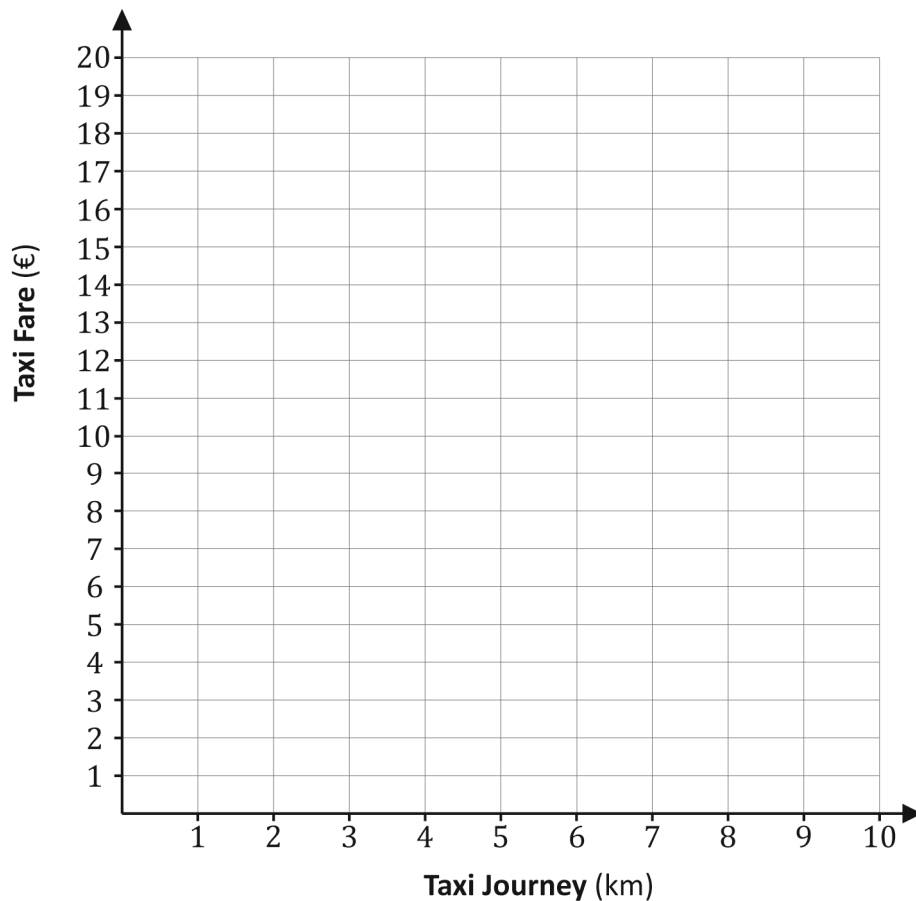


- (a)** Complete the table below to show the taxi fare for each of the journeys listed. Five are already done for you.

Taxi Journey (km)	1	2	3	4	5	6	7	8	9	10
Taxi Fare (€)	3.50			8		11		14		17

[illegible]

- (b)** Draw a graph on the diagram below showing the cost of a taxi fare for all journeys of between 1 km and 10 km.



- (c)** Use your graph from part **(b)** to answer the following questions. Show your work on the graph.

- (i) Grace needs to travel 5 km to get home. Write down her taxi fare in euro.

Taxi fare = €

- (ii) James has €15.50. Write down how far James can travel by taxi on his budget.

Taxi journey = km

- (d)** Describe in your own words how to calculate the cost of a taxi fare for a journey of 15 km.

[illegible]

- (e) Write a formula to represent the cost of a taxi fare for a journey any distance.

[illegible]

- (f)** Use your formula from part **(e)** to verify your answer in part **(c)(i)**.

A large grid of graph paper with 20 columns and 10 rows. The grid is composed of small squares, with a thicker vertical line separating the first column on the left from the rest of the grid.

Question 4 (Suggested maximum time: 10 minutes)

Question 4 (Suggested maximum time: 10 minutes)

The number of minutes that a footballer played in each game over a league season were observed and are listed below.

63,	59,	67,	90,	80,
90,	77,	81,	84,	55,
74,	90,	68,	59,	61,
87,	51,	51,	79,	90.



- (a)** Work out how many games the footballer played over the season.

[illegible]

- (b)** Complete the stem and leaf diagram below to show this data.

5							
6							
7							
8							
9							

Key:

5

9

$$=$$

- (c) Write down the **mode** of the data.

Mode =

- (d)** Work out the **mean** of the data.
Give your answer correct to the nearest minute.

[illegible]

- (e)** Work out the percentage of games in which the footballer played 80 minutes or more. Give your answer correct to the nearest whole number.

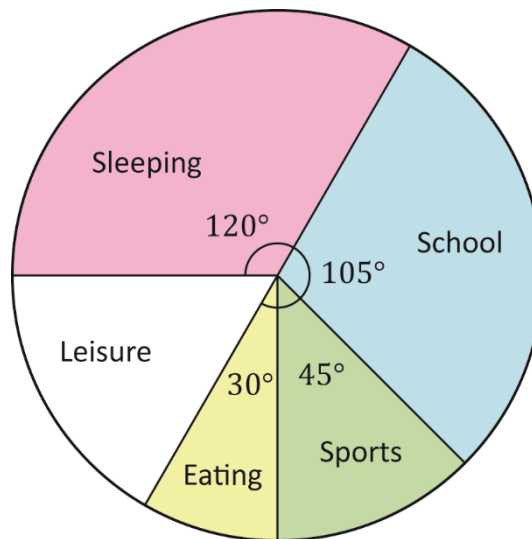
[illegible]

Question 5 (Suggested maximum time: 5 minutes)

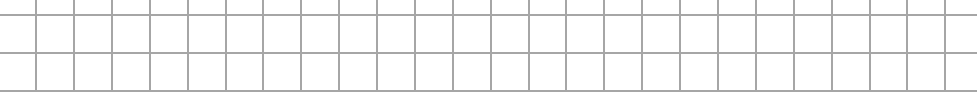
Question 5 (Suggested maximum time: 5 minutes)

John recorded all of the different activities he did over 24 hours on a given day.

The results are shown in the pie chart below.



- (a)** Work out the size of the angle that represents Leisure activities.

A large rectangular area filled with a light gray grid pattern, intended for drawing or writing. The grid consists of small squares, approximately 20 units wide and 20 units high.

- (b)** What activity did John spend the greatest amount of time on during that day? Give a reason for your answer.

Answer: _____

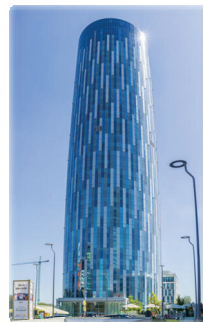
Reason: _____

- (c)** How many hours did John spend on Sports during that day?

Question 6

(Suggested maximum time: 10 minutes)

The Floreasca City Sky Tower is a famous cylindrical building in Romania's capital city, Bucharest.



The Floreasca City Sky Tower is 137 m tall.

There are 32 floors above ground level.

- (a) Tick (✓) the correct box to show which value best represents the average height of each floor of the Sky Tower.

Tick **one** box only. Justify your answer.

3.05 m

☐

3.78 m

☐

4.28 m

☐

Justification:

- (b) The roof of the building is flat, in the shape of a circle.

The length of the circumference of the roof is 40π m.

Tick (✓) the correct box to show the radius length of the roof.

Tick **one** box only. Justify your answer.

40 m

☐

20 m

☐

10 m

☐

Justification:

Hint: $2 \times \pi \times r = 40\pi$

- (c) Work out the volume of the Floreasca City Sky Tower.

Give your answer correct to the nearest m^3 .

$V = \pi \times r^2 \times h$

Question 7**(Suggested maximum time: 10 minutes)**

- (a) Simplify $7a + 12b + 2a - 4b$.

- (b) Multiply out and simplify $(3x + 2)(x + 2)$.

- (c) Lisa says that $x = 2$ is a solution of $x^2 - 12x + 20 = 0$. Show that Lisa is correct.

- (d) Solve the following simultaneous equations:

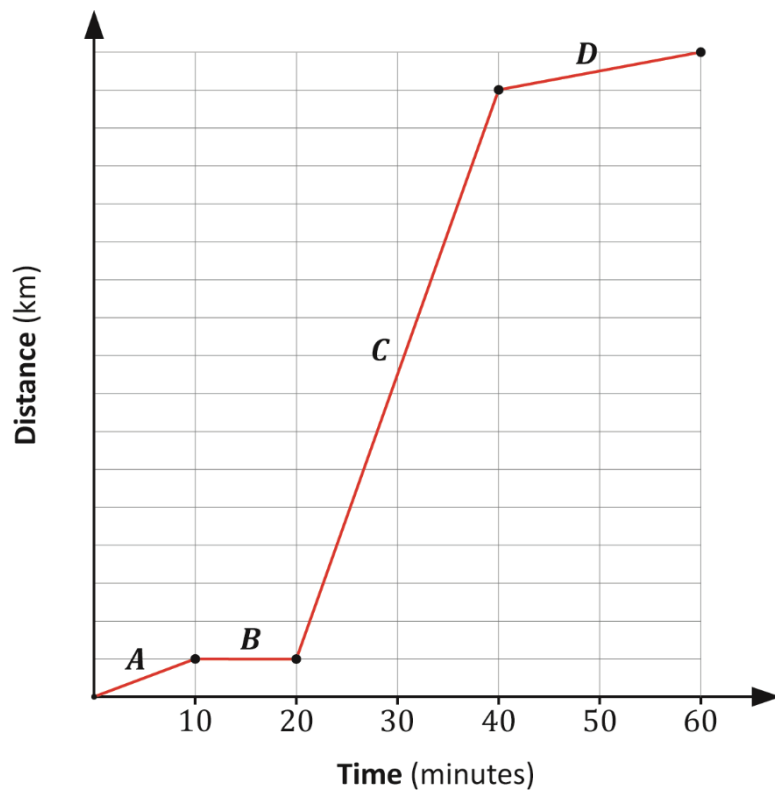
$$\begin{aligned}x + y &= 16 \\x - y &= -10.\end{aligned}$$

Question 8 (Suggested maximum time: 5 minutes)

Question 8 (Suggested maximum time: 5 minutes)

Ava takes the bus to visit her grandparents. The graph below shows the distance in kilometres travelled by Ava after t minutes during her journey.

The graph is in four sections, labelled **A**, **B**, **C** and **D**.



- (a) Write the correct letter, **A**, **B**, **C** or **D**, into the table to match each description to the correct section of the graph.

Description	Section of the graph (A, B, C or D)
Ava waits 10 minutes for the bus.	
Ava walks for 20 minutes.	
Ava walks for 10 minutes and stops.	
Ava takes the bus for 20 minutes.	

- (b)** The bus travels at an average speed of 45 km/hr.
Find the distance that Ava travelled, in km, on the bus.



Question 9

(Suggested maximum time: 5 minutes)

- (a) Jordan rolls a fair die.

For each result in the table below, use **one** of the terms:

Impossible, Unlikely, Equal chance, Likely or Certain

to describe the probability that it happens.



Result	Probability
Jordan rolls a 6.	
Jordan rolls an even number.	
Jordan rolls a number greater than 2.	
Jordan rolls a number less than 1.	

- (b) Fernando plays a game where he flips a fair coin and then rolls a fair die.

- (i) Complete the two-way table below to show all of the possible outcomes for Fernando.

	1	2	3	4	5	6
H	H, 1					
T				T, 4		

- (ii) Write down the probability that Fernando will get a head and a 6.

Answer:

- (iii) Write down the probability that Fernando will get a tail and an even number.

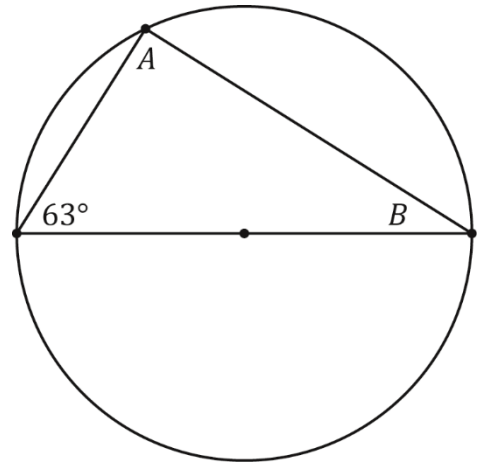
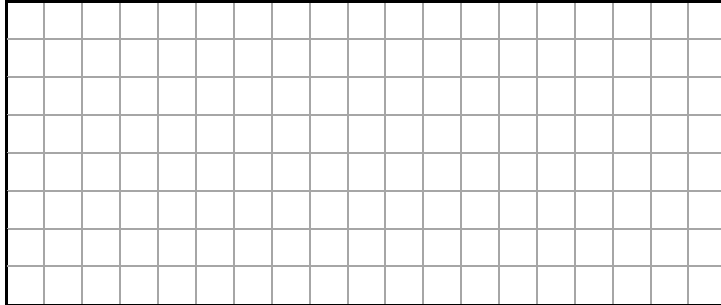
Answer:

Question 10

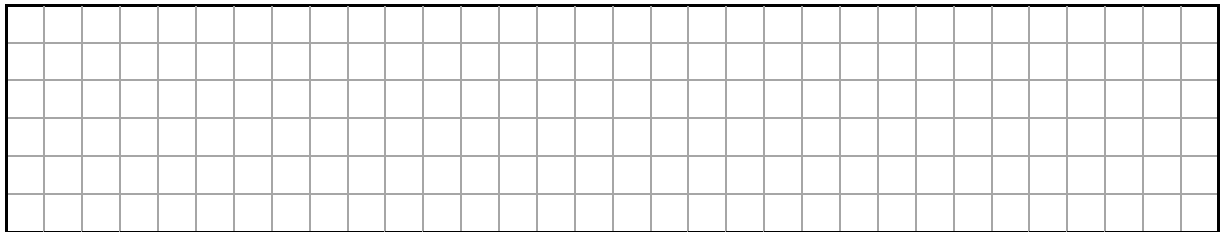
(Suggested maximum time: 10 minutes)

- (a) The triangle shown has an angle of size 63° and sits on the diameter of the circle.
(diagram not to scale)

- (i) Give a reason why $\angle A$ is an angle of size 90° .

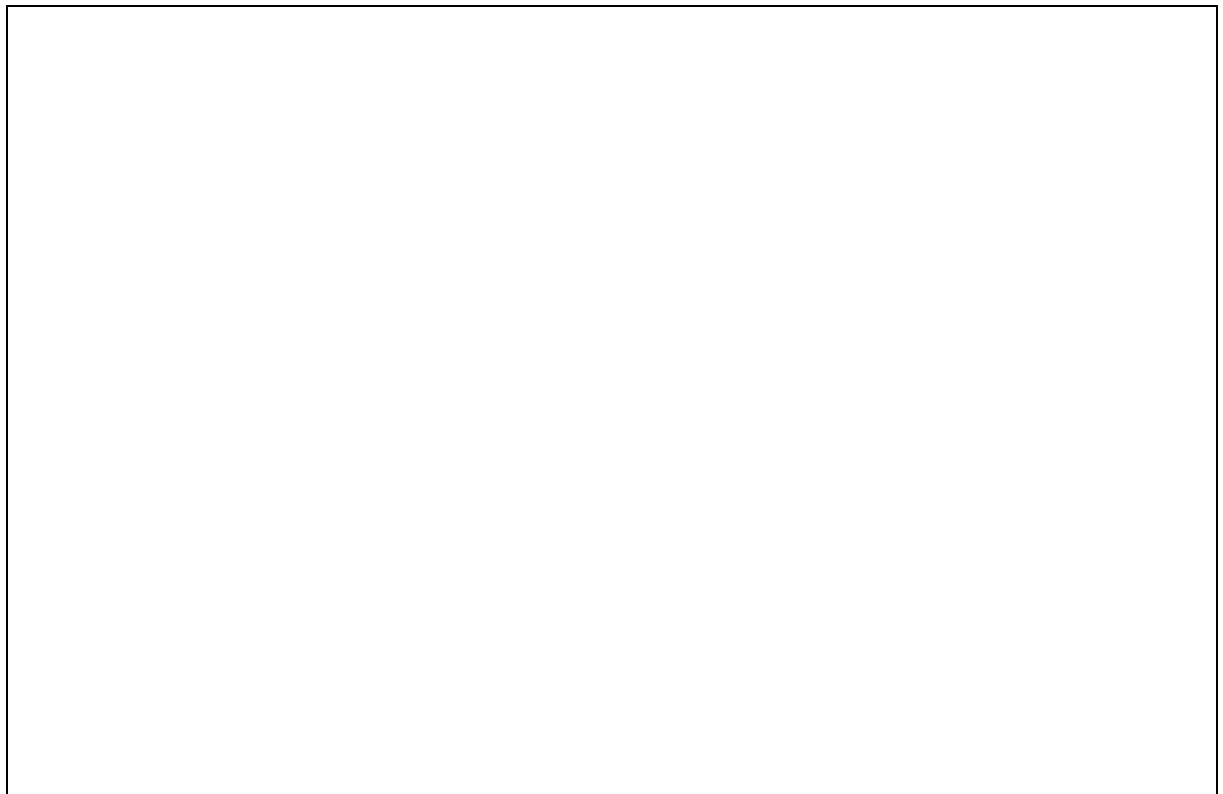


- (ii) Work out the size of the angle B , the third angle in the triangle.



- (b) Draw in the space below, a line segment $[AB]$, where $|AB| = 12$ cm.

Divide $[AB]$ into **three equal** segments without measuring it.
Show all of your construction lines clearly.



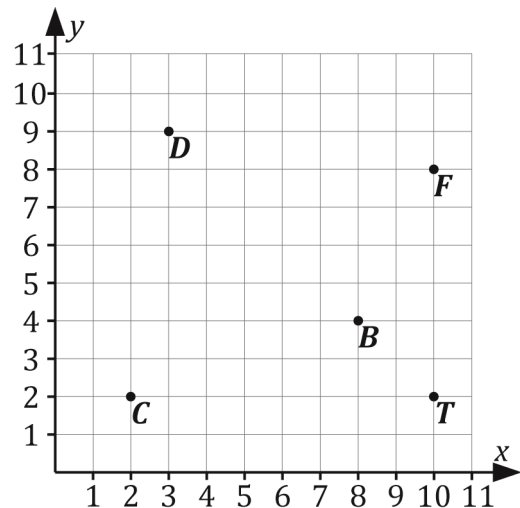
Question 11

(Suggested maximum time: 10 minutes)

A science student discovered various flowers in a field while on an ecology field trip.

The field is laid out in a grid and the position of each flower is shown.

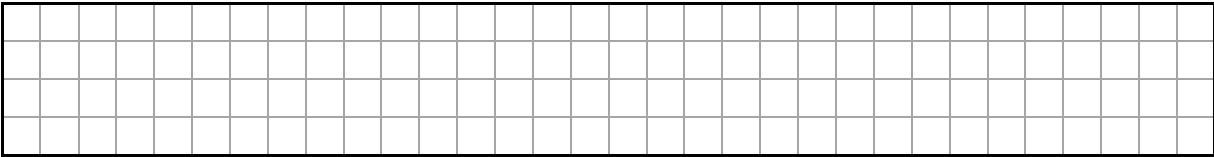
The flowers found are as follows:
Buttercup (B), Foxglove (F),
Daisy (D), Thistle (T)
and a Cowslip (C).



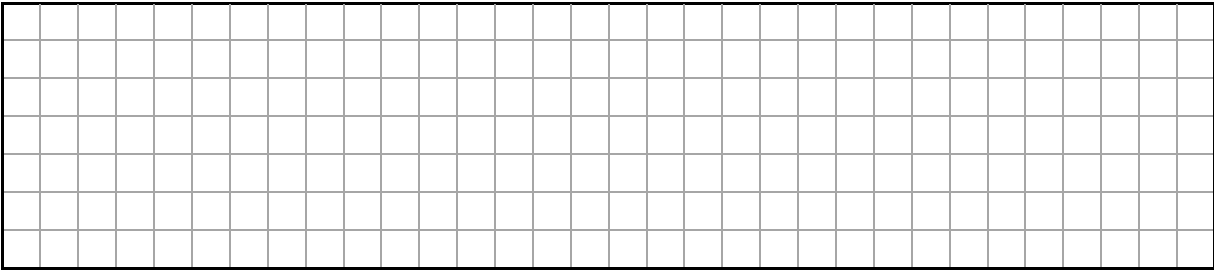
- (a) Complete the table below to show the co-ordinates of the position of each flower. One is done for you.

Point	B	F	D	T	C
Co-ordinates	(,)	(10, 8)	(,)	(,)	(,)

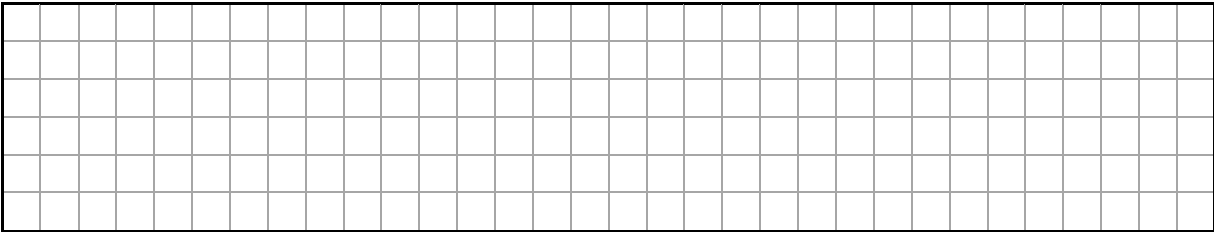
- (b) Each square represents 1 m². Find the total area of the field.



- (c) Find the distance between the cowslip (C) and the foxglove (F) in the field.

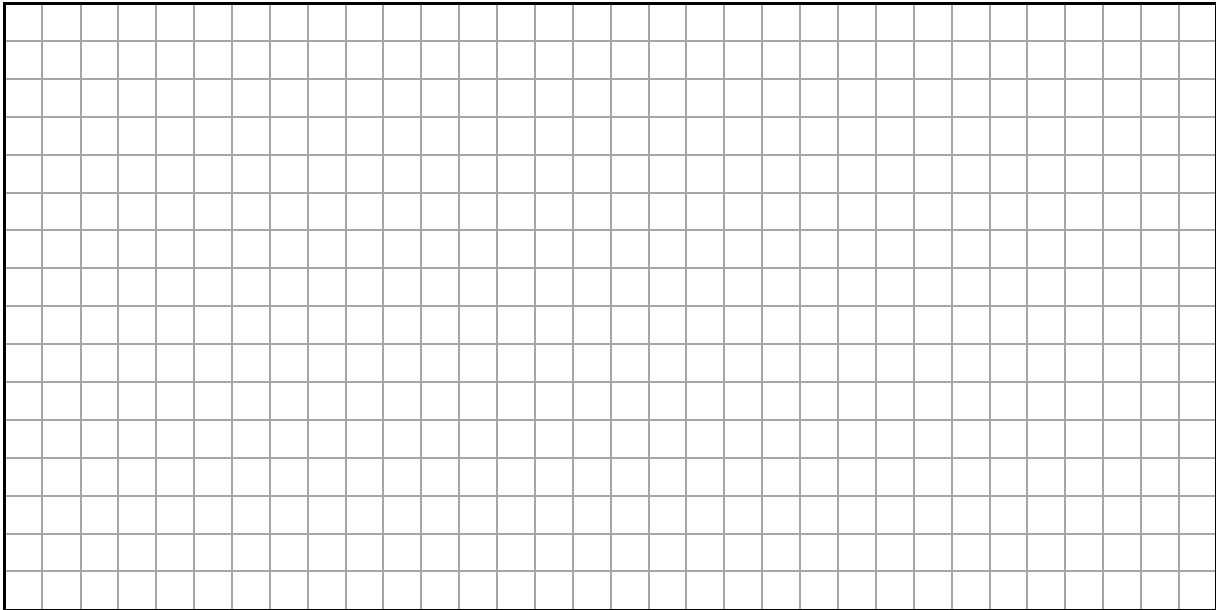


- (d) What is the slope of the line joining the buttercup (B) to the thistle (T) on the grid?



(a) Complete the table below for the function $f(x) = x^2 + 2x + 1$ for $-4 \leq x \leq 1, x \in \mathbb{R}$.

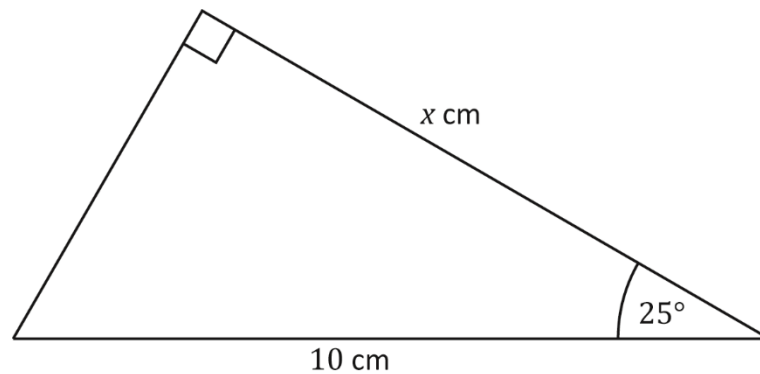
x	$f(x)$
-4	
-3	
-2	
-1	
0	
1	



Question 13 (Suggested maximum time: 5 minutes)

Question 13 (Suggested maximum time: 5 minutes)

The triangle below is a right-angled triangle with a hypotenuse of length 10 cm and one angle of size 25° . (diagram not to scale)



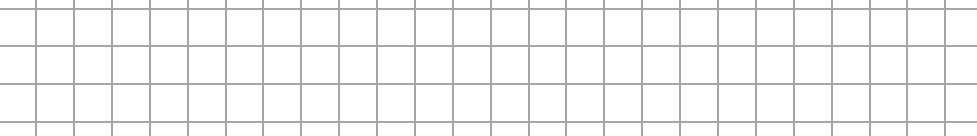
- (a)** Using the diagram, write $\cos 25^\circ$ as a fraction in terms of x .

$\cos 25^\circ = \frac{\text{adjacent}}{\text{hypotenuse}} =$	
---	--

- (b)** Use a calculator to find $\cos 25^\circ$, correct to one decimal place.

[illegible]

- (c)** Use your answers from part **(a)** and part **(b)** to work out the value of x .



Page for extra work.

Label any extra work clearly with the question number and part.

[illegible]

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Pre-Junior Cycle Final Examination, 2023

Mathematics – Ordinary Level

Time: 2 hours

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